OurMocks

2027 CYCLE

MOCK · PRACTICE · CRACK

CUET UG

Mathematics

Integration and its Applications

Antiderivatives, Definite Integrals and Area · Section A 3 + B1 III + B2 III · CUET UG 2027 Cycle Notes

INSIDE
Concept Notes

INSIDE
Solved Examples

INSIDE
Practice MCQs + Solutions

Aligned to the latest official syllabus · Concept-first explanations

www.ourmocks.com

Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) 2026 syllabus — Mathematics/Applied Mathematics (319), the current 2027 baseline)	Skills the questions test
Indefinite integrals and evaluation	"Indefinite integrals of simple functions. Evaluation of indefinite integrals." (Section A Unit 3, compulsory) — "Integration of a variety of functions by substitution, by partial fractions and by parts, evaluation of simple integrals of the following types and problems based on them" (Section B1 Mathematics Unit III)	Picking the right method in seconds; carrying $+C$ without being reminded.
Standard special integrals	The syllabus's named integral family: $\frac{1}{x^2+a^2}$, $\frac{1}{\sqrt{x^2 \pm a^2}}$, $\sqrt{a^2-x^2}$ and the quadratic-completing forms. (Section B1 Mathematics Unit III)	Recognising the pattern after one algebraic nudge; completing the square on demand.
Definite integrals and properties	"Fundamental Theorem of Calculus (without proof). Basic properties of definite integrals and evaluation of definite integrals." (Section B1 Mathematics Unit III)	Evaluating $F(b)-F(a)$ cleanly; deploying King's rule, additivity and symmetry.
Area under simple curves	"Applications in finding the area under simple curves, especially lines, circles/parabolas/ellipses (in standard form only)." (Section B1 Mathematics Unit III; Section A Unit 3 phrases it as "Application of integration as area under the curve (simple curve)")	Root–Split–Flip discipline; upper-minus-lower setups with verified positioning.

EXAM PATTERN

CUET UG pattern (NTA official information bulletin, 2026 cycle — the current 2027 baseline): each subject paper carries **50 compulsory questions** in **60 minutes**, with +5 for every correct answer and −1 for every incorrect answer. The 2026 syllabus lists this unit as **Section A Unit 3 (compulsory for every candidate)** and inside **Section B1 Unit III** and **Section B2 Unit III** — integration is where the paper trades computation speed for marks, so every technique below is paired with its fastest route.

Indefinite Integrals — Differentiation in Reverse

An antiderivative of f is any function F whose derivative is f: $F'(x) = f(x)$. The indefinite integral $\int f(x)\,dx = F(x) + C$ collects the **whole family**, because derivatives destroy constants — x^2, $x^2 + 5$ and $x^2 - \pi$ all differentiate to $2x$, so the antiderivative must carry a $+C$. That constant is not decoration; questions grade it, and its presence signals you understand "indefinite" means **not pinned to one curve**. The verification habit is free and unbeatable: differentiate your answer and check you land back on the integrand.

THE STANDARD FORMS — YOUR MULTIPLICATION TABLES

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1), \qquad \int \frac{dx}{x} = \ln|x| + C$$

$$\int e^x\,dx = e^x + C, \quad \int a^x\,dx = \frac{a^x}{\ln a} + C$$

$$\int \sin x\,dx = -\cos x + C, \quad \int \cos x\,dx = \sin x + C$$

$$\int \sec^2 x\,dx = \tan x + C, \quad \int \operatorname{cosec}^2 x\,dx = -\cot x + C$$

- **Linearity does the housekeeping.** Sums split term by term, and constants slide out front: $\int (3x^2 + 2\sin x)\,dx = x^3 - 2\cos x + C$.
- **The power rule's one exception.** $\int x^{-1}\,dx$ would divide by zero — nature's way of introducing $\ln|x|$, with the modulus because the antiderivative must also work for negative x.
- **Signs alternate on trig pairs.** sin → −cos, cos → sin: differentiate each and check — the flipped sign of the sine integral is the oldest distractor in the chapter.

WORKED EXAMPLE — A BASELINE SET, VERIFIED

Evaluate $\int \left(4x^3 - \frac{6}{x} + 2\sin x\right) dx$.

Solution. Term by term: $4 \cdot \frac{x^4}{4} = x^4$; $-6\ln|x|$; $2(-\cos x) = -2\cos x$. So

$$\int \left(4x^3 - \frac{6}{x} + 2\sin x\right) dx = x^4 - 6\ln|x| - 2\cos x + C.$$

Verify by differentiating: $4x^3 - \frac{6}{x} + 2\sin x$ — exactly the integrand. Twenty seconds, zero risk. Most "hard" MCQs in this family are three of these wearing a coat.

The Four Methods — Ask in Order

Every integrand answers one question: **which method?** Ask them in a fixed order and the decision makes itself. (1) Is it a **standard form** already? Write the answer. (2) Is an **inner function's derivative sitting outside** the main expression? Substitution. (3) Is it **rational with a factorable denominator**? Partial fractions. (4) Is it a **product of two different families**

(polynomial with exponential, polynomial with trig, x with $\ln x$)? By parts, with ILATE choosing u. This order is not a slogan — it runs from cheapest to most expensive, exactly how you want to spend exam minutes.

Substitution. If you spot $g'(x)$ multiplied near $g(x)$, substitute $u = g(x)$, $\mathrm{d}u = g'(x)\,\mathrm{d}x$. The integral collapses to a standard form in u, you integrate, then substitute back. Worked: $\int \frac{2x}{x^2+1}\,\mathrm{d}x$ — the numerator $2x$ is precisely the derivative of x^2+1, so with $u = x^2+1$ the integral becomes $\int \frac{\mathrm{d}u}{u} = \ln|u| + C = \ln(x^2+1) + C$ (no modulus needed since $x^2+1 > 0$ always). Trig flavour: $\int \sin^2 x \cos x\,\mathrm{d}x$ puts $u = \sin x$, $\mathrm{d}u = \cos x\,\mathrm{d}x$, giving $\frac{u^3}{3} + C = \frac{\sin^3 x}{3} + C$.

Partial fractions. A rational function with a factorable denominator splits into a sum of simpler fractions, each of which integrates to a log or arctan. Worked: $\int \frac{\mathrm{d}x}{x^2-1}$ — write $\frac{1}{(x-1)(x+1)} = \frac{1}{2}\left[\frac{1}{x-1} - \frac{1}{x+1}\right]$, so the integral is $\frac{1}{2}\ln\left|\frac{x-1}{x+1}\right| + C$. The setup takes one line: cover up $(x-1)$ and evaluate the rest at $x = 1$ to get the coefficient $\frac{1}{2}$, then cover $(x+1)$ and evaluate at $x = -1$ to get $-\frac{1}{2}$.

By parts. $\int uv\,\mathrm{d}x = u\int v\,\mathrm{d}x - \int\left(u'\int v\,\mathrm{d}x\right)\mathrm{d}x$ — one family is integrated (v), the other differentiated (u). **ILATE** ranks which is u: Inverse trig, Logarithmic, Algebraic, Trigonometric, Exponential — earlier in the list wins u. Worked: $\int xe^x\,\mathrm{d}x$ — algebraic beats exponential, so $u = x$, $v = e^x$:

$$\int xe^x\,\mathrm{d}x = xe^x - \int e^x\,\mathrm{d}x = (x-1)e^x + C.$$

And $\int \ln x\,\mathrm{d}x$ has no visible partner — take $u = \ln x$, $v = 1$: the answer $x\ln x - x + C$ is the by-parts trick used on an "empty" product, and it appears constantly.

EVERYDAY EXAMPLE — CHOOSING THE RIGHT KITCHEN TOOL

You do not chop vegetables with a rolling pin. Each job picks its tool by **shape**: a peeler for skin, a grinder for paste, a whisk for air. Integration is the same triage — the integrand's shape (naked standard form? derivative pair? fraction? two-family product?) names the tool before any computation starts. Students who "try things randomly" are cooking with the rolling pin; students who run the four-question order finish the method **before** the algebra begins. The decision flow diagram at the end of this chapter is the kitchen wall chart — pin it there.

The Special Standard Integrals

The syllabus names a family of integrals that are **standard forms by decree** — you memorise them the way you memorised 7×8. They all come from inverse trig or log antiderivatives, and questions wrap them in one algebraic nudge (usually completing the square) before they fit.

THE NAMED FAMILY

$$\int \frac{\mathrm{d}x}{x^2+a^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C \qquad \int \frac{\mathrm{d}x}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C$$

$$\int \frac{\mathrm{d}x}{x^2-a^2} = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + C$$

$$\int \frac{\mathrm{d}x}{\sqrt{x^2+a^2}} = \ln\left|x+\sqrt{x^2+a^2}\right| + C$$

$$\int \sqrt{a^2-x^2}\,\mathrm{d}x = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) + C$$

- **Completing the square is the universal adapter.** $\int \frac{\mathrm{d}x}{x^2+4x+13}$ becomes $\int \frac{\mathrm{d}x}{(x+2)^2+3^2}$ — the arctan form with $a=3$, answer $\frac{1}{3}\tan^{-1}\left(\frac{x+2}{3}\right)+C$. One shift, one scale, done.
- **The quarter-circle integral is worth knowing cold:** $\int_0^a \sqrt{a^2-x^2}\,\mathrm{d}x = \frac{\pi a^2}{4}$ — it literally computes a quarter of a circle's area, and questions lean on this geometric reading repeatedly.
- **Match by silhouette.** Denominator quadratic with **no** x-term in the numerator: arctan or log family. Square root of a quadratic: arcsin or the `sqrt` family. The numerator's shape picks the branch.

Definite Integrals and the Fundamental Theorem

A definite integral has limits, and the Fundamental Theorem of Calculus turns it into subtraction: $\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a)$, where F is **any** antiderivative. No $+C$ survives — it would cancel between the two evaluations. Upper limit first, lower second, in that order; reversing them flips the sign. The definite integral is a **number**, not a function, and that single idea powers every property below.

PROPERTIES THAT SAVE WHOLE MINUTES

$$\int_a^b f = -\int_b^a f \qquad \int_a^b f = \int_a^c f + \int_c^b f$$

$$\int_0^a f(x)\,\mathrm{d}x = \int_0^a f(a-x)\,\mathrm{d}x \quad \text{(King's rule)}$$

$$\int_{-a}^{a} f = 0 \text{ if } f \text{ is odd}, \quad 2\int_0^a f \text{ if } f \text{ is even}$$

- **King's rule reflects the interval.** On $[0,a]$, replacing x by $a-x$ often turns an ugly integrand into a twin; adding the two forms can make the integrand collapse to a constant. This single trick decides entire questions in two lines.
- **Symmetry is a free elimination.** Over $[-a,a]$, odd integrands (x^3, $\sin x$, $x\cos x$) integrate to exactly 0 — no computation, just recognition.
- **Additivity splits at trouble points.** An absolute value $|x-1|$ or a piecewise definition inside the integrand: split the interval at the joint and integrate each piece with its honest sign.

WORKED EXAMPLE — KING'S RULE DOING THE HEAVY LIFTING

Evaluate $\int_0^{\pi} \frac{x\sin x}{1+\cos^2 x}\,\mathrm{d}x$.

Solution. King's rule with $a = \pi$: replace x by $\pi - x$. Since $\sin(\pi - x) = \sin x$ and $\cos^2(\pi - x) = \cos^2 x$, the transformed integral is

$$\int_0^{\pi} \frac{(\pi - x)\sin x}{1 + \cos^2 x}\,\mathrm{d}x.$$

Add the original and the twin: $2I = \int_0^{\pi} \frac{\pi \sin x}{1+\cos^2 x}\,\mathrm{d}x$ — the x terms cancel. Now substitute $u = \cos x$, $\mathrm{d}u = -\sin x\,\mathrm{d}x$, limits $1 \to -1$:

$$2I = \pi \int_{-1}^{1} \frac{\mathrm{d}u}{1+u^2} = \pi\big[\tan^{-1} u\big]_{-1}^{1} = \pi\Big(\frac{\pi}{4} - \Big(-\frac{\pi}{4}\Big)\Big) = \frac{\pi^2}{2},$$

so $I = \frac{\pi^2}{4}$. Without King's rule this integral is a monster; with it, four lines.

Area Under Simple Curves

Area is the definite integral's most visual application — with one non-negotiable discipline. The integral computes **signed** area: regions below the x-axis come in negative. The question "area bounded by the curve and the axis" wants **geometric** area, always positive. The workflow is **Root–Split–Flip**: find where the curve meets the axis, split the interval at those roots, integrate each piece, and take absolute values of below-axis pieces before adding.

- **Line under a parabola:** the region between $y = x^2$ and $y = 4$ runs from $x = -2$ to $x = 2$; upper minus lower gives $\int_{-2}^{2}(4 - x^2)\,\mathrm{d}x = \big[4x - \frac{x^3}{3}\big]_{-2}^{2} = \frac{32}{3}$ square units. Always verify **which curve is on top** — here the line, at every point of the interval.
- **Circle reading:** $y = \sqrt{a^2 - x^2}$ is the upper semicircle of radius a; its integral over $[-a, a]$ is $\frac{\pi a^2}{2}$, and over $[0, a]$ a quarter — matching the special integral exactly. Geometry confirms calculus; calculus confirms geometry.
- **Absolute values inside the integrand** are pre-split signals: for $\int_0^3 |x - 1|\,\mathrm{d}x$, split at $x = 1$ and add the two honest pieces: $\int_0^1 (1 - x)\,\mathrm{d}x + \int_1^3 (x - 1)\,\mathrm{d}x = \frac{1}{2} + 2 = \frac{5}{2}$.

WORKED EXAMPLE — PARABOLA CUT BY A LINE, END TO END

Find the area of the region bounded by the parabola $y = x^2$ and the line $y = 4$.

Solution. Intersections: $x^2 = 4$ gives $x = \pm 2$. On $[-2, 2]$ the line $y = 4$ sits above $y = x^2$ (test $x = 0$: $4 > 0$). Upper minus lower:

$$A = \int_{-2}^{2} (4 - x^2)\,\mathrm{d}x = \left[4x - \frac{x^3}{3}\right]_{-2}^{2} = \left(8 - \frac{8}{3}\right) - \left(-8 + \frac{8}{3}\right) = \frac{32}{3}.$$

The even integrand lets you halve the work: $2\int_0^2 (4 - x^2)\,\mathrm{d}x$ — same answer, one sign-chart risk fewer. The answer is a number with square units; if yours is negative, the subtraction order was backwards.

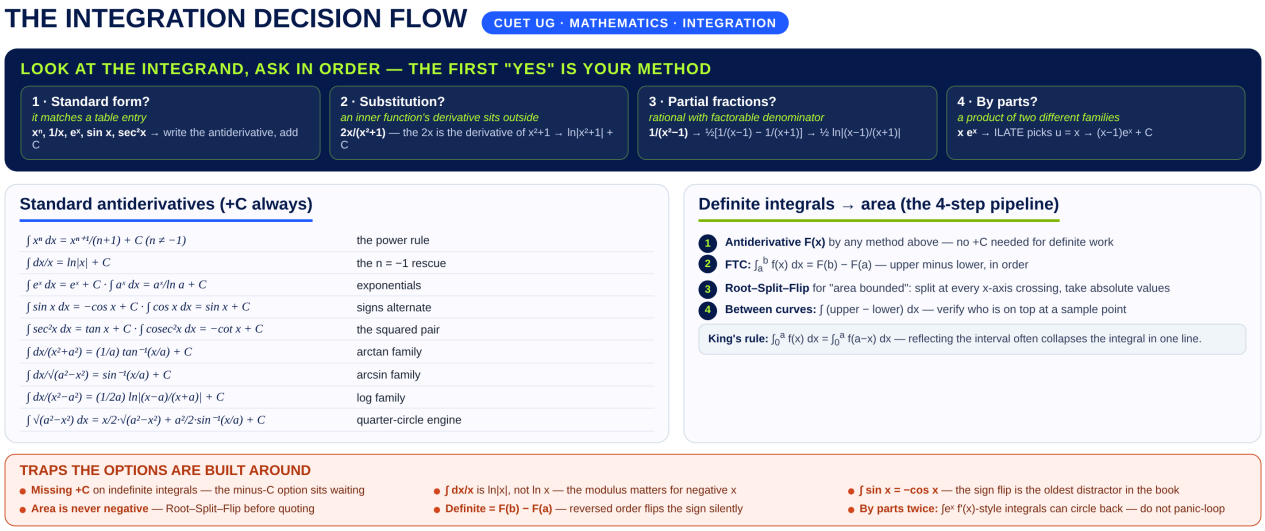


Figure 1: The decision flow on one board: four methods in asking order, the standard-form table, and the four-step area pipeline.

Formula Cheatsheet — the Whole Chapter on One Page

FORMULA CHEATSHEET — INTEGRATION AND ITS APPLICATIONS

Formula / rule	Letters mean	Fire it when
$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C,\ n \neq -1$	raise power, divide	any naked power
$\int \frac{dx}{x} = \ln\lvert x\rvert + C$	modulus included	the $n = -1$ rescue
$\int e^x = e^x + C,\ \int a^x = \frac{a^x}{\ln} a + C$	exponential family	growth forms
$\int \sin x = -\cos x + C,\ \int \cos x = \sin x + C$	signs alternate	trig basics
$\int \sec^2 x = \tan x + C$	squared pair with cosec	trig basics
Substitution: $u = g(x),\quad du = g'(x)\,dx$	inner derivative visible	derivative pairs
Partial fractions: split $\frac{1}{(x-a)(x-b)}$	cover-up coefficients	factorable rationals
By parts: u by ILATE, v the easy integral	In Log Alg Trig Exp	two-family products

Formula / rule	Letters mean	Fire it when
$\int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$	complete the square first	quadratic bottoms
$\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C$	arc family	root denominators
$\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\ln\left\lvert\frac{x-a}{x+a}\right\rvert + C$	log family	difference squares
FTC: $\int_a^b f = F(b) - F(a)$	upper minus lower	all definite integrals
King: $\int_0^a f(x) = \int_0^a f(a-x)$	reflect and add	trig-weighted limits
Symmetry: odd $\to 0$ on $[-a, a]$, even $\to$ double	recognition beats work	symmetric limits
Area: Root-Split-Flip; between curves: upper − lower	verify who is on top	area questions

Common Errors and Special Cases

COMMON ERRORS AND SPECIAL CASES — INTEGRATION

The mistake	What is right	The trap it sets
Dropping $+C$ on indefinite integrals	Every indefinite answer carries the constant	Options both with and without C — one is always bait
Writing $\ln x$ for $\int \frac{dx}{x}$	$\ln\lvert x\rvert$ — negative x needs the modulus	Modulus-free distractor
Sign-slipping $\int \sin x$	$-\cos x + C$	The positive twin sits right beside it
Definite integrals carrying $+C$	It cancels in $F(b) - F(a)$	An option awkwardly displaying C
Arctan vs log on quadratics	Denominator + shape: arctan; − shape: log	Swapped-family options
Quoting raw negative integrals as area	Root-Split-Flip first	Negative-value area options
Upper-lower subtraction reversed	$F(b) - F(a)$, always	Sign-flipped numeric answers
Ignoring the joint in $\lvert x-1\rvert$	Split at the zero of the inside	Options from integrating $x - 1$ straight through

The mistake	What is right	The trap it sets
ILATE ignored	Inverse-trig and Log outrank Algebraic for u	By-parts answers that loop without resolving

Sixty-Second Self-Test

Statement	Verdict
$\int e^x \, dx = e^x + C$.	True — the self-deriving function
A definite integral should carry $+C$.	False — it cancels between limits
$\int_{-5}^{5} x^3 \, dx = 0$.	True — odd function, symmetric limits
An area bounded by a curve can be negative.	False — geometric area is never negative
$\int \frac{dx}{x^2+a^2} = \ln(x^2 + a^2) + C$.	False — it is the arctan family
King's rule applies on $[0, a]$ by reflecting $x \to a - x$.	True — the reflect-and-add trick

OurMocks

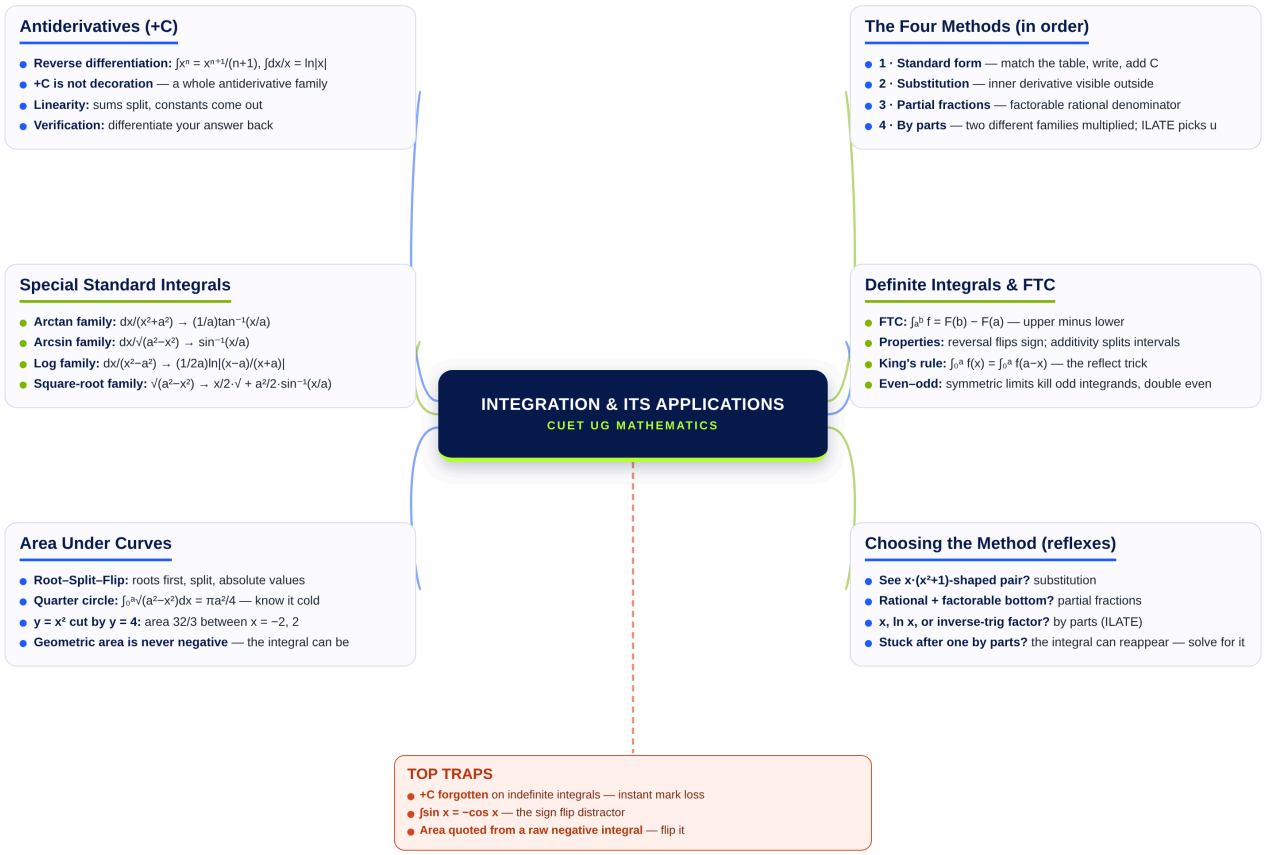


Figure 2: The chapter in one look: antiderivatives, four methods, special integrals, FTC properties and the area pipeline.

Practice Questions — CUET UG Pattern (10 MCQs)

Ten questions in the CUET UG style. Each correct answer earns +5; each incorrect one costs −1. Run the decision flow before any algebra — the method choice **is** the question.

Q1. $\int \left(3x^2 + \frac{2}{x}\right) \mathrm{d}x$ equals:

(A) $x^3 + 2\ln|x| + C$ (B) $x^3 + 2\ln x$
(C) $x^3 - 2\ln|x| + C$ (D) $3x^3 + 2\ln|x| + C$

Q2. $\int \sin x \, \mathrm{d}x$ equals:

(A) $\cos x + C$ (B) $-\sin x + C$
(C) $-\cos x + C$ (D) $\sin x + C$

Q3. $\int_0^{\frac{\pi}{2}} \sin^2 x \, \mathrm{d}x$ equals:

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$
(C) 1 (D) π

Q4. $\int \frac{dx}{x^2+9}$ equals:

(A) $\tan^{-1}\left(\frac{x}{3}\right)+C$ **(B)** $\frac{1}{9}\tan^{-1}\left(\frac{x}{9}\right)+C$
(C) $\frac{1}{6}\tan^{-1}\left(\frac{x}{3}\right)+C$ **(D)** $\frac{1}{3}\tan^{-1}\left(\frac{x}{3}\right)+C$

Q5. $\int xe^x\,dx$ equals:

(A) xe^x+C **(B)** $(x-1)e^x+C$
(C) $(x+1)e^x+C$ **(D)** $\frac{x^2}{2}\cdot e^x+C$

Q6. $\int_1^e \frac{dx}{x}$ equals:

(A) 1 **(B)** $e-1$
(C) e **(D)** 2

Q7. The area bounded by the parabola $y=x^2$ and the line $y=4$ is:

(A) $\frac{16}{3}$ **(B)** 16
(C) $\frac{64}{3}$ **(D)** $\frac{32}{3}$

Q8. $\int_{-2}^{2} x^3\,dx$ equals:

(A) 8 **(B)** 4
(C) 0 **(D)** 16

Q9. $\int \frac{dx}{x^2-4}$ equals:

(A) $\frac{1}{2}\ln|x^2-4|+C$ **(B)** $\frac{1}{4}\ln\left|\frac{x-2}{x+2}\right|+C$
(C) $\frac{1}{4}\ln\left|\frac{x+2}{x-2}\right|+C$ **(D)** $\frac{1}{2}\tan^{-1}\left(\frac{x}{2}\right)+C$

Q10. $\int_0^3 |x-1|\,dx$ equals:

(A) 3 **(B)** $\frac{9}{2}$
(C) $\frac{5}{2}$ **(D)** $\frac{7}{2}$

Detailed Solutions

Sol. Q1. Split: $\int 3x^2\,dx = x^3$ and $\int \frac{2}{x}\,dx = 2\ln|x|$. Total $x^3+2\ln|x|+C$. Option (B) drops the modulus and the constant together — a two-fault distractor.

Answer: (A)

Sol. Q2. Differentiate each option and match: $(-\cos x)' = \sin x$. The sign alternation is the whole question — $\cos x$ differentiates to $-\sin x$, not $\sin x$.

Answer: (C)

Sol. Q3. Standard result: $\int_0^{\frac{\pi}{2}} \sin^2 x\,\mathrm{d}x = \frac{\pi}{4}$ (from $\sin^2 x = \frac{1-\cos 2x}{2}$: the integral is $\frac{x}{2} - \sin 2\frac{x}{4}$ on $[0, \frac{\pi}{2}]$, giving $\frac{\pi}{4}$). Memorise the half-square value — it recurs in half the trig-weighted questions.

Answer: (B)

Sol. Q4. Arctan family with $a = 3$: $\int \frac{\mathrm{d}x}{x^2+a^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C = \frac{1}{3}\tan^{-1}\left(\frac{x}{3}\right) + C$. Option (A) forgot the $\frac{1}{a}$ scale, option (B) scaled both places.

Answer: (D)

Sol. Q5. By parts, ILATE: algebraic x outranks exponential e^x, so $u = x$, $v = e^x$: $xe^x - \int e^x\,\mathrm{d}x = (x-1)e^x + C$. Option (C) is what appears when e^x is chosen as u — the ILATE rule exists to prevent exactly that wrong first move.

Answer: (B)

Sol. Q6. $[\ln|x|]_1^e = \ln e - \ln 1 = 1 - 0 = 1$. The cleanest definite integral in the chapter — a one-line recall check.

Answer: (A)

Sol. Q7. Intersections at $x = \pm 2$; the line sits above the parabola (test $x = 0$). $A = \int_{-2}^{2}(4 - x^2)\,\mathrm{d}x = \left[4x - \frac{x^3}{3}\right]_{-2}^{2} = \frac{32}{3}$. The even integrand halves the work if you prefer $2\int_0^2$.

Answer: (D)

Sol. Q8. Odd integrand over symmetric limits $[-2, 2]$: the left lobe and right lobe cancel exactly, so the answer is 0 with no computation. Symmetry recognition is the entire question.

Answer: (C)

Sol. Q9. Partial fractions: $\frac{1}{(x-2)(x+2)} = \frac{1}{4}\left[\frac{1}{x-2} - \frac{1}{x+2}\right]$, integrating to $\frac{1}{4}\ln\left|\frac{x-2}{x+2}\right| + C$. Option (C) has the fraction inverted; option (A) is the "integrate 1/u" reflex misapplied.

Answer: (B)

Sol. Q10. Split at the zero of the inside: $\int_0^1 (1-x)\,\mathrm{d}x + \int_1^3 (x-1)\,\mathrm{d}x = \left[x - \frac{x^2}{2}\right]_0^1 + \left[\frac{x^2}{2} - x\right]_1^3 = \frac{1}{2} + 2 = \frac{5}{2}$. Integrating $x - 1$ straight through gives $\frac{5}{2} - \frac{9}{2} = -2$… the sign says a piece was flipped — exactly why the split comes first.

Answer: (C)

Wrong-Answer Clinic

If you picked this	The thinking slip	The lasting fix
(B) in Q1	Accepted the answer without the modulus or $+C$	Two-point checklist on every indefinite: `abs` where logs appear, C always
(A) in Q2	Trusted the "cos pairs with sin" sound	Differentiate the option back — verification is faster than doubt
(A) in Q4	Arctan formula without the $\frac{1}{a}$ scale	Write the full standard form before substituting a
(D) in Q5	Matched shapes instead of running by parts	ILATE picks $u = x$; one by-parts pass settles it mechanically
(C) in Q9	Inverted the log fraction	Cover-up method fixes coefficients mechanically — trust the signs
(B) in Q10	Integrated through the corner	Absolute value = split at the root of the inside, always

© OurMocks · www.ourmocks.com · CUET UG Mathematics · Integration and its Applications · 2027 cycle